KAPASA MAKASA UNIVERSITY

FACULTY OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY

ISB 140/DS 110 – INTRODUCTION TO STATISTICS

Tutorial sheet 4

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1. (a) Find z for each of the following confidence levels.

   (i) 90% (ii) 95% (iii) 96% (iv) 97% (v) 98% (vi) 99%

   (b) Briefly explain the meaning of each of the following terms.

   (i) Null hypothesis (ii) Alternative hypothesis (iii) Critical point(s)

   iv) Significance level (v) non-rejection region (vi) Rejection region (vii) Tails of a test (viii) Two types of errors

   (c) Explain which of the following is a two-tailed test, a left-tailed test, or a right-tailed test.

   (i) $H_0: \mu = 12,\ H_1: \mu < 12$ (ii) $H_0: \mu \leq 85,\ H_1: \mu > 85$

   (iii) $H_0: \mu = 33,\ H_1: \mu \neq 33$

   (d) Show the rejection and non-rejection regions for each of these cases by drawing a sampling distribution curve for the sample mean, assuming that it is normally distributed.

2. Find the appropriate rejection regions for the large-sample test statistic z in these cases:

   (a) A right-tailed test with a $\alpha = 0.01$

   (b) A two-tailed test at the 5% significance level

   (c) A left-tailed test at the 1% significance level

   (d) A two-tailed test with a $\alpha = 0\ .01$

3. A random sample of 40 farmers gave a sample mean of $\bar{X} = k860$ received by farmers per 100 pounds of watermelon. Assume that σ is known to be $k240$ per 100 pounds.
   (a) Find a 90% confidence interval for the population mean price (per 100lbs) that farmers get for their watermelon crop. What is the margin of error?
   (b) Find a 99% confidence interval. Do you expect a longer or shorter interval for a higher level of confidence?
4. A random sample of $n = 35$ observations from a quantitative population produced a mean $\bar{X} = 2.4$ and a standard deviation $S = 0.29$. Suppose your research objective is to show that the population mean μ exceeds 2.3
   (a) Give the null and alternative hypotheses for the test.
   (b) Locate the rejection region for the test using a 5% significance level.
   (c) Find the standard error of the mean.
   (d) Before you conduct the test, use your intuition to decide whether the sample mean $\bar{X}$ 2.4 is likely or unlikely, assuming that $\mu = 2.3$. Now conduct the test. Do the data provide sufficient evidence to indicate that $\mu = 2.3$?
5. Find the value of *t* for the *t* distribution for each of the following.
   (a) Area in the right $tail = .05$ and $df = 12$
   (b) Area in the left $tail = .025$ and $n = 66$
   (c) Area in the left $tail = .001$ and $df = 49$
   (d) Area in the right $tail = .005$ and $n = 24$
6. For a population, the value of the standard deviation is 2.65. A random sample of 35 observations taken from this population produced the following data.
   42 51 42 31 28 36 49 29 46 37 32 27 33 41 47 41 28 46 34 39 48 26 35 37 38 46 48 39 29 31 44 41 37 38 46
   (a) What is the point estimate of μ?
   (b) Make a 98% confidence interval for μ.
   (c)What is the margin of error of estimate for part (b)?

7. By using the critical value approach, fill in the blanks in the given table below

Test Statistics	Significance level	One or two tailed tests	Critical value	Rejection region	conclusion
$z^* = 0.88$	$\alpha = 0.05$	Two-tailed			
$z^* = -2.67$	$\alpha = 0.05$	One-tailed (lower)			
$z^* = 5.05$	$\alpha = 0.01$	Two-tailed			
$z^* = -1.22$	$\alpha = 0.01$	One-tailed (lower)			

8. Determine the sample size for the estimate of μ for the following:

(a) $e = 2.3, \sigma = 15.40, \text{confidence level} = 99\%$

(b) $e = 4.1, \sigma = 23.45, \text{confidence level} = 95\%$

(c) $e = 25.9, \sigma = 122.25, \text{confidence level} = 90\%.$

(d) $e = .17, \sigma = .90, \text{confidence level} = 99\%$

(e) $e = 1.45, \sigma = 5.82, \text{confidence level} = 95\%$

(f) $e = 5.65, \sigma = 18.20, \text{confidence level} = 90\%$

9. Test $H_0 : \sigma_1^2 = \sigma_2^2$ versus $H_1 : \sigma_1^2 \neq \sigma_2^2$ for the following data.

$n_1 = 10;\ \bar{X} = 71;\ S_1^2 = 64$ and $n_2 = 25;\ \bar{X} = 131;\ S_2^2 = 96$:

10. (a) Suppose that a February Gallup poll of 1200 randomly selected voters found that 53 Percent support George W. Bush's energy policy. Conduct a hypothesis test at a level of significance of $\alpha = 0.01$ to test whether the true voter population support for George W. Bush's energy policy in February was less than 56 Percent.

(b) Report the $P-$value of the test in (a) and give a practical interpretation of it.

11. For a data set obtained from a random sample, $n = 81$ and $\bar{X} = 48.25$. It is known that $\sigma = 4.8$.

(a) What is the point estimate of μ?

(b) Make a 95% confidence interval for μ.

(c) What is the margin of error of estimate for part (b)?

12. (a) The standard deviation for a population is $\sigma = 14.8$. A random sample of 25 observations selected from this population gave a mean equal to 143.72. The population is known to have a normal distribution.

(i) Make a 99% confidence interval for μ.

(ii) Construct a 95% confidence interval for μ.

(iii) Determine a 90% confidence interval for μ.

(iv) Does the width of the confidence intervals constructed in parts a through c decrease as the confidence level decreases? Explain your answer.

(b) The standard deviation for a population is $\sigma = 7.14$. A random sample selected from this population gave a mean equal to 48.52.

(i) Make a 95% confidence interval for μ assuming $n = 196$.

(ii) Construct a 95% confidence interval for μ assuming $n = 100$.

(iii) Determine a 95% confidence interval for μ assuming $n = 49$.

(iv) Does the width of the confidence intervals constructed in parts a through c increase as the sample size decreases? Explain.

13. A reading test is given to both a control group and an experimental group (which received special tutoring). The average score for the 30 subjects in the control group was 349.2 with a sample standard deviation of 56.6. The average score for the 30 subjects in the experimental group was 368.4 with a sample standard deviation of 39.5. Use a 4% level of significance to test the claim that the experimental group performed better than the control group.

14. (a) For a population data set, $\sigma = 12.5$.

(i) How large a sample should be selected so that the margin of error of estimate for a 99% confidence interval for μ is 2.50?

(ii) How large a sample should be selected so that the margin of error of estimate for a 96% confidence interval for μ is 3.20?

(b) For a population data set, $\sigma = 14.50$.

(i) What should the sample size be for a 98% confidence interval for μ to have a margin of error of estimate equal to 5.50?

(ii) What should the sample size be for a 95% confidence interval for μ to have a margin of error of estimate equal to 4.25?

15. A vendor was concerned that a soft drink machine was not dispensing 6 ounces per cup, on average. A sample size of 40 gave a mean amount per cup of 5.95 ounces and a standard deviation of 0.15 ounce, again assume that we know $\sigma = 0.15$

(a) Find the $P - value$ for a hypothesis test to determine if the mean is different from 6 ounces

(b) For which of the following levels of significance would the null hypothesis be rejected? (i) $\alpha = 0.10$ (ii) $\alpha = 0.05$ (iii) $\alpha = 0.04$ (iv) $\alpha = 0.01$

(c) For each case in part (b), what type of error has possibly been committed?

(d) Find a 96% confidence interval for the mean amount of soda dispensed per cup.

(e) Is your interval in (d) consistent with the test conclusion in (b) (iii)? Explain.

(f) Based on your answer to (b) (iv) would you expect a 99% confidence interval to contain 6?

(g) Suppose that the population standard deviation is $\sigma = 0.15$, what sample size would be needed so that the maximum error in a 96% confidence interval is $e = .01$?

16. Suppose you wish to find the average annual salary for school teachers in the United States. Suppose you took a random sample and found that $\bar{X} = 53,145$ and you estimated $\sigma = \$16,451$.

(a) Find a 95% confidence interval for the mean salary, given that your sample had size (i) $n = 36$, (ii) $n = 64$, (iii) $n = 144$.

(b) Does the margin of error increase or decrease as n gets larger. What effect did quadrupling n have on the margin of error?

(c) Does the confidence interval get longer or shorter as the sample size increases? Is that what you would expect?

17. You wish to test a claim that the average time to complete a form is 15 minutes. You believe it is longer. For this, you surveyed a random sample of 45 people and found that it took them an average of 17 minutes to complete the form with a standard deviation of 5 minutes which you have reason to believe is the population standard deviation. Conduct your test with a level of significance of $\alpha = 0.05$.

18. The lifetimes of light bulbs produced by a particular manufacturer have a mean of 1200 hours and a standard deviation of 400 hours. The population distribution is normal. Suppose

that you purchase nine bulbs, which can be regarded as a random sample from the manufacturer's output.

(a) What is the mean of the sample mean lifetime?
(b) What is the variance of the sample mean?
(c) What is the standard error of the sample mean?
(d) What is the probability that, on average, those nine light bulbs have lives of fewer than 1050 hours?

19. Find z_c where $c =$ critical value for (a) $c = 0.99$, (b) $c = 0.95$ and (c) $c = 0.92$. (d) $c = 0.90$.

20. In a survey of 1000 large corporations, 250 said given a choice between a job candidate who smokes and an equally qualified non-smoker, the non-smoker would get the job

(a) Let p represent the proportion of all corporations preferring a non-smoking candidate. Find a point estimate for p.
(b) Find a 95% confidence interval for p.
(c) How would a news writer report these results? What was the margin of error for the 95% confidence interval?

21. A random sample of $n_1 = 288$ voters registered in the state of California showed that 141 voted in the last general election. A random sample of $n_2 = 216$ registered voters in Colorado showed that 125 voted in the last general election. Do these data indicate that the population proportion of voter turnout in Colorado is higher than that in California? Use a 5% level of significance.

22. It was reported that the average life span of people in Hawaii is 77 years. Conduct a hypothesis test to see if the life span of people in Honolulu is less than 77 years (Hint: use both the critical value approach and the p-value approach). Use a level of significance of $\alpha = 0.05$. For this test, a random sample of 20 obituary notices of people in Honolulu had a mean $\bar{X} = 71.4$ years, and $S \approx 20.65$.

23. A sample of 37 adult male desert bighorn sheep indicated the standard deviation of the sheep weights to be $15.8\ lb$.

(a) How many adult male sheep should be included in a sample to be 90% sure that the sample mean weight $\bar{X}$ is within $2.5\ lb$ of the population mean weight μ for all such bighorn in the region.
(b) Assuming you have found an error in the original calculation and the standard deviation was $25.8\ lb$. How many adult male sheep should be included in a sample to

be 90% sure that the sample mean weight $\bar{X}$ is within 2.5 lb of the population mean weight μ for all such bighorn in the region.

24. A developer wishes to test whether the mean depth of water below the surface in a large development tract was less than 500 feet. The sample data was as follows: $n = 32$ test holes, the sample mean was 486 feet, and the standard deviation was 53 feet, assume that this is the population standard deviation $\sigma = 53$. Complete the test by computing the $P - value$, and report the conclusion for a 1% level of significance.

25. A new diet and exercise program has been advertised as a remarkable way to reduce blood glucose levels in diabetic patients. Ten randomly selected diabetic patients are put on the program, and the results after 1 month are given by the following table:

Before	268	225	252	192	307	228	246	298	231	185
After	106	186	223	110	203	101	211	176	194	203

Do the data provide sufficient evidence to support the claim that the new program reduces blood glucose level in diabetic patients? Use $\alpha = 0.05$.

26. In a 1999 survey of 80 Computer Science graduates and 110 Electrical Engineering graduates, it was found that the Computer Science graduates had a mean starting salary of k48100 with a standard deviation of k7200, while the Electrical Engineering graduates had a mean starting salary of k52900 with a standard deviation of k5300.

(a) Find a point estimate for the difference in average starting salaries for Computer Science and Electrical Engineering graduates.

(b) Let μ_1 be the population mean starting salary for the Computer Science graduates and let μ_2 be the population mean salary for the Electrical Engineering graduates. Find a 96% confidence interval for $\mu_1 - \mu_2$. Assume the standard deviations given are the population standard deviations.

(c) Based on the interval in (b), would you be comfortable saying that the mean starting salary for Computer Science graduates is less than that for Electrical Engineering graduates? Explain.

(d) How would your answer change if the standard deviations given are sample standard deviations, and you were asked to find a 95% confidence interval?

27. For healthy adults, the mean blood pH is $\mu = 7.4$. It is suspected a new arthritis drug changes blood pH (either higher or lower). Test this at a 5% significance level. To do

this, a sample of 31 patients on the new drug were tested, and the sample had a mean $\bar{X} = 8.1$ and a standard deviation $S = 1.9$

28. (a) A random sample of size 20 is drawn from a population having a normal distribution. The sample mean and the sample standard deviation from the data are given, respectively, as $\bar{X} = 2.2$ and $S = 1.42$. Construct a 90% CI for the population variance σ^2 and interpret.

(b) A medicine is suspected of causing an elevated heart rate in a certain group of high-risk patients. Twenty patients from the group were given the medicine. The changes in heart rates were found to be as follows:

-1 8 5 10 2 12 7 9 1 3

4 6 4 12 11 2 -1 10 2 8

Construct a 95% CI for the variance of change in heart rate. Assume that the population has a normal distribution and interpret.

29. (a) A study was conducted to see the differences between oxygen consumption rates for male runners from a college who had been trained by two different methods, one involving continuous training for a period of time each day and the other involving intermittent training of about the same overall duration. The means, standard deviations, and sample sizes are shown in the following table:

Continuous training: $n_1 = 15$, $\bar{X}_1 = 46.28$ and $S_1 = 6.3$

Intermittent training: $n_2 = 7$, $\bar{X}_2 = 42.34$ and $S_2 = 7.8$

If the measurements are assumed to come from normally distributed populations with equal variances, estimate the difference between the population means, with confidence coefficient 0.95, and interpret.

(b) The following information was obtained from two independent samples selected from two normally distributed populations with unknown but equal variances:

Sample: 1 14 15 12 13 6 14 11 12 17 19 23

Sample: 2 16 18 12 20 15 19 15 22 20 18 23 12 20

Construct a 95% CI for the difference between the population means and interpret.

30. Two independent samples from a normal population are taken with the following summary statistics:

$$n_1 = 16, \qquad \bar{X}_1 = 2.4, \qquad S_1 = 0.1$$

$$n_2 = 11, \qquad \bar{X}_2 = 2.6, \qquad S_2 = 0.5$$

Construct a 95% CI for σ_1^2/σ_2^2.

31. A university employment office wants to compare the time taken by graduates with three different majors to find their first fulltime job after graduation. The following table lists the time (in days) taken to find their first full-time job after graduation for a random sample of eight business majors, seven computer science majors, and six engineering majors who graduated in May 2014.

Business	**Computer science**	**Engineering**
208	156	126
162	113	275
240	281	363
180	128	146
148	305	298
312	147	392
176	232	
292		

At a 5% significance level, can you reject the null hypothesis that the meantime taken to find their first full-time job for all May 2014 graduates in these fields is the same?

32. Suppose that in an attempt to determine whether a die is fair, you tossed it 120 times with the following results:

Outcome	1	2	3	4	5	6
Frequency	24	18	15	26	21	16

Does this provide sufficient evidence at a level of significance of $\alpha = .05$ to conclude that the die is not fair? Be sure to find the test statistic, P-value and state the conclusion.

33. Home Mail Corporation sells products by mail. The company's management wants to find out if the number of orders received at the company's office on each of the 5 days of the week is the same. The company took a sample of 400 orders received during a 4-

week period. The following table lists the frequency distribution for these orders by the day of the week.

Day of the week	Monday	Tuesday	Wednesday	Thursday	Friday
Number of orders received	92	71	65	83	89

Test at a 5% significance level whether the null hypothesis that the orders are evenly distributed over all days of the week is true.

34. A large company buys thousands of light bulbs every year. The company is currently considering four brands of light bulbs to choose from. Before the company decides which light bulbs to buy, it wants to investigate if the mean lifetimes of the four types of light bulbs are the same. The company's research department randomly selected a few bulbs of each type and tested them. The following table lists the number of hours (in thousands) that each of the bulbs in each brand lasted before being burned out.

Brand I	**Brand II**	**Brand III**	**Brand IV**
23	19	23	26
24	23	27	24
19	18	25	21
26	24	26	29
22	20	23	28
23	22	21	24
25	19	27	28

At a 5% significance level, test the null hypothesis that the mean lifetime of bulbs for each of these four brands is the same.

35. Lack of discipline has become a major problem in schools in the United States. A random sample of 300 adults was selected, and these adults were asked if they favor giving more freedom to schoolteachers to punish students for lack of discipline. The two-way classification of the responses of these adults is presented in the following table.

	In favor (F)	Against (A)	No opinion (N)

Men (M)	93	70	12
Women (W)	87	32	6

(a) Calculate the expected frequencies for this table, assuming that the two attributes, gender and opinions on the issue, are independent

(b) Does the sample provide sufficient evidence to conclude that the two attributes, gender and opinions of adults, are dependent? Use a 1% significance level.

36. The following data represent a random sample of end-of-year bonuses for lower-level managerial personnel employed by a large firm. Bonuses are expressed in percentage of yearly salary.

Female	6.2	9.2	8.0	7.7	8.4	9.1	7.4	6.7
Male	8.9	10.0	9.4	8.8	12.0	9.9	11.7	9.8

The objective is to determine whether the male and female bonuses are t same. We can answer this question by connecting the following:

(a) Use the ANOVA approach to test the appropriate hypothesis. Use a $\alpha = 0.05$.

(b) What assumptions are necessary for the test in (a)?

(c) Test the appropriate hypothesis by using the two-sample t-test for comparing population means. Compare the value of the t −statistic with the value of the F −statistic calculated in (a).

37.

	Population 1	Population 2	Population 3	Population 4
Row 1	24	81	60	121
Row 2	46	64	91	72
Row 3	20	37	105	93

(a) Write the null and alternative hypotheses for a test of homogeneity for this table.

(b) Calculate the expected frequencies for all cells assuming that the null hypothesis is true.

(c) For α = .025, find the critical value of χ^2. Show the rejection and non-rejection regions on the chi-square distribution curve.

(d) Find the value of the test statistic χ^2.

(e) Using α = .025, would you reject the null hypothesis?